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Studierichting: Informatica
Oefeningenreeks 4A nr. 47.2

$$\int \frac{3x-1}{(x^2+4x+5)^3} dx$$
$$x^2+4x+5 = (x+2)^2+1$$

$$3x-1 = \frac{3}{2}(2x+4) - 7$$

Dus:

$$\int \frac{3x-1}{(x^2+4x+5)^3} dx$$
$$= \frac{3}{2} \int \frac{2x+4}{(x^2+4x+5)^3} dx - 7 \int \frac{dx}{((x+2)^2+1)^3}$$

Eerste integraal:

$$u = x^2 + 4x + 5$$

$$du = (2x+4)dx$$

$$\frac{3}{2} \int u^{-3} du = \frac{3}{2} \left(\frac{u^{-2}}{-2} \right)$$
$$= -\frac{3}{4u^2}$$
$$= -\frac{3}{4(x^2+4x+5)^2}$$

Tweede integraal:

$$x+2 = \tan t$$

$$dx = \sec^2 t dt$$

$$((x+2)^2+1)^3 = (\tan^2 t + 1)^3 = (\sec^2 t)^3 = \sec^6 t$$

Dus:

$$-7 \int \frac{\sec^2 t}{\sec^6 t} dt$$
$$= -7 \int \cos^4 t dt$$

Gebruik:

$$\cos^4 t = \left(\frac{1 + \cos(2t)}{2} \right)^2$$

$$\begin{aligned}
&= \frac{1}{4}(1 + 2\cos(2t) + \cos^2(2t)) \\
&= \frac{1}{4}\left(1 + 2\cos(2t) + \frac{1 + \cos(4t)}{2}\right) \\
&= \frac{3}{8} + \frac{1}{2}\cos(2t) + \frac{1}{8}\cos(4t)
\end{aligned}$$

Dus:

$$\begin{aligned}
&-7 \int \cos^4 t dt \\
&= -7 \int \left(\frac{3}{8} + \frac{1}{2}\cos(2t) + \frac{1}{8}\cos(4t)\right) dt \\
&= -7 \left(\frac{3}{8}t + \frac{1}{4}\sin(2t) + \frac{1}{32}\sin(4t)\right) \\
&= -\frac{21}{8}t - \frac{7}{4}\sin(2t) - \frac{7}{32}\sin(4t)
\end{aligned}$$

Terug substitueren:

$$t = \text{Bgtan}(x + 2)$$

$$\begin{aligned}
\sin(2t) &= \frac{2 \tan t}{1 + \tan^2 t} \\
&= \frac{2(x + 2)}{(x + 2)^2 + 1} \\
&= \frac{2(x + 2)}{x^2 + 4x + 5} \\
\cos(2t) &= \frac{1 - \tan^2 t}{1 + \tan^2 t} \\
&= \frac{1 - (x + 2)^2}{(x + 2)^2 + 1} \\
\sin(4t) &= 2 \sin(2t) \cos(2t) \\
&= \frac{4(x + 2)(1 - (x + 2)^2)}{((x + 2)^2 + 1)^2} \\
&= \frac{4(x + 2)(1 - (x + 2)^2)}{(x^2 + 4x + 5)^2}
\end{aligned}$$

Dus:

$$\begin{aligned}
&\int \frac{3x - 1}{(x^2 + 4x + 5)^3} dx \\
&= -\frac{3}{4(x^2 + 4x + 5)^2} - \frac{21}{8} \text{Bgtan}(x + 2) \\
&\quad - \frac{7(x + 2)}{2(x^2 + 4x + 5)} - \frac{7(x + 2)(1 - (x + 2)^2)}{8(x^2 + 4x + 5)^2} + C
\end{aligned}$$

Vereenvoudigen geeft:

$$\boxed{-\frac{7x+17}{4(x^2+4x+5)^2} - \frac{21(x+2)}{8(x^2+4x+5)} - \frac{21}{8} \operatorname{Bgtan}(x+2) + C}$$

References